AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Assignment Title:	Test plan and test case		
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Course Title:	Software Engineering		
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Semester:	Spring 2024-25	Course Teacher:	**TONNY SHEKHA KAR**

Group Name/No.: **03**

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Faculty use only		
FACULTY COMMENTS	**Marks Obtained**	
	Total Marks	

Test Planning

Testing Categories , their Objectives & test cases:

1. Unit Testing

A) Objective:

- To validate that individual functions, methods, and classes work correctly in isolation.
- Ensures that each core component (e.g., login validator, anomaly detection rule) behaves as expected.

B) Test Cases:

1. Validate correct login credentials.
2. Reject incorrect password during login.
3. Test OTP generation function for MFA.
4. Verify function for anomaly score calculation.
5. Unit test DDoS detection algorithm logic with simulated traffic patterns.
6. Validate phishing URL detection and alert generation mechanism
7. Validate the SQL injection detection algorithm function.

2. Integration Testing

A) Objective:

- To verify that different modules and services work together as intended.
- Tests interaction between the login system, anomaly engine, and notification services.

B) Test Cases:

1. Login module integration with anomaly detection.
2. Login + MFA system – successful authentication flow.
3. Verify database updates after an alert is triggered.
4. UI to backend.
5. Integration with dashboard display module.(dashboard UI with real-time updates)
6. Trigger DDoS → activate mitigation or alert system.
7. Integrate phishing detection status in system dashboard.
8. Integrate SQL injection scanner with user input fields

3. Black-box Testing

A) Objective:

- To validate the system's functionality from a user's perspective without accessing the source code.
- Focuses on inputs and expected outputs (e.g., login failure for wrong password).

B) Test Cases:

1. Submit valid login and receive dashboard access.
2. Enter invalid email format – show error message.
3. Try login without internet – system handles timeout.
4. Input valid but unused email – shows "User not found."

5. Use expired MFA code – system blocks access.

6. Try submitting a form with empty fields – validation blocks.

7. Check report generation for a single anomaly event.

4. White-box Testing

A) Objective:

- To assess internal logic, decision structures, loops, and code paths.

- Ensures thorough coverage of control structures in detection algorithms and logic modules.

B) Test Cases:

1. Execute all logic branches in the login validator.

2. Test all conditions in the anomaly detection rule set.

3. Decision tree for SQL pattern matching – all paths.

4. Full path coverage of phishing detection logic.

5. Check all branches in report formatting logic.

6. Verify error handling path when DB connection fails.

7. Logic coverage for MFA code expiration and retry.

5. System Testing

A) Objective:

- To evaluate the complete CyberLayer system as a whole in a simulated real-world environment.

- Ensures all subsystems integrate to meet the overall requirements.

B) Test Cases:

1. Full login → MFA → anomaly trigger → report generation flow.
2. User signs up, logs in, and uses the system without error.
3. Verify logs are recorded for each interaction.
4. Check system behavior after 100+ concurrent logins.
5. User triggers 3 attack types – full report generated.
6. System shutdown + restart → data integrity preserved.
7. Login failure from multiple devices triggers an alert.
8. Try exporting reports in PDF – correct format.
9. All detection modules can run simultaneously.
10. Verify UI reflects correct status of attack modules real time.

Test Cases

Project Name: CyberLayer		Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_1		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Login Session		Test Execution date:
Test Title: Validate correct login credentials		
Description: Test website login page		
Precondition: User must have valid username and password		
Test Steps	Test Data	Expected Results
1.Go to the website 2.Enter valid username 3.Enter valid password 4.Click submit	*Username: user@example *Password: CorrectPassword123	1.User is redirected to the dashboard 2.Login session is created and stored in session table
Post Condition: User is validated with database and successfully logs into account. The account session details are logged in the database.		

Project Name: CyberLayer		Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_2		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Login Session		Test Execution date:
Test Title: Reject login with incorrect password		
Description: Ensure system prevents login with wrong password		
Precondition: User exists with a known username		
Test Steps	Test Data	Expected Results
1.Go to the website 2.Enter valid username 3.Enter valid password 4.Click submit	*Username: user@example *Password: CorrectPassword123	1.Error message: "Invalid password" 2.User remains on login page 3.Attempt is logged
Post Condition: Login attempt is rejected and recorded as failed		

Project Name: CyberLayer		Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_3		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Login Session		Test Execution date:
Test Title: Validate OTP generation for MFA		
Description: Verify the generation of One-Time Password for second-factor authentication		
Precondition: User is authenticated with correct credentials		
Test Steps	Test Data	Expected Results
1. Login with valid username and password 2. System triggers OTP generation 3.Send OTP via email/SMS	Phone: +1234567890 or Email: user@example.com	1.OTP is generated 2.OTP is delivered to user 3.OTP is logged with expiry timestamp
Post Condition: OTP is generated and sent to user securely		

Project Name: CyberLayer	Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_4	Test Designed date:13.05.25
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Anomaly Detection	Test Execution date:

Test Title: Test anomaly score calculation

Description: Ensure function calculates score based on behavioral metrics

Precondition: User behavior data available (e.g. location, time, device)

Test Steps	Test Data	Expected Results
1. Simulate login behavior 2. Call anomaly detection function 3. Capture calculated score	*Login attempt with mismatched location or unusual time	1. Anomaly score is generated 2. If score exceeds threshold, system flags it as suspicious

Post Condition: Score is returned and compared to threshold

Project Name: CyberLayer		Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_5		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: DDoS detection		Test Execution date:
Test Title: Validate DDoS detection algorithm		
Description: Confirm algorithm correctly identifies distributed traffic anomalies		
Precondition: Simulated traffic data is available		
Test Steps	Test Data	Expected Results
1.Simulate burst traffic from multiple IPs 2.Run DDoS detection algorithm 3.Check alert mechanism	*IP burst traffic pattern (e.g. 1000 requests/sec from 100 IPs)	1. DDoS alert is triggered 2. Log entry for attack attempt is created
Post Condition: DDoS flag is raised		

Project Name: CyberLayer		Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_6		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Phishing		Test Execution date:
Test Title: Detect phishing URL on login input		
Description: Ensure login system detects phishing or malicious URLs		
Precondition: Ensure login system detects phishing or malicious URLs		
Test Steps	Test Data	Expected Results
1.Submit input with suspicious URL 2. System evaluates URL 3. Generate alert if matched	URL: http://fake-login.example.com	1.Input is rejected 2.Alert message generated 3.Admin is notified
Post Condition: Input is blocked, and admin is notified		

Project Name: CyberLayer		Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_7		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: SQLi detection		Test Execution date:
Test Title: Detect SQL injection on login input		
Description: Ensure SQLi inputs are blocked and logged		
Precondition: System includes SQLi detection function		
Test Steps	Test Data	Expected Results
1.Enter SQL injection string in username/password 2. Submit login form 3. System parses input	Input: `' OR 1=1 --`	1.Injection detected 2.Login attempt denied 3.Log entry created
Post Condition: Input is flagged and login is denied		

Project Name: CyberLayer		Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_8		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Authentication + Anomaly Engine		Test Execution date:
Test Title: Verify integration of login module with anomaly detection		
Description: Ensure that login attempts are evaluated by anomaly detection system		
Precondition: Anomaly detection module must be active		
Test Steps	Test Data	Expected Results
1.Attempt login with valid credentials from unusual location/device 2.Login module sends data to anomaly detection system 3.Anomaly system evaluates and responds	1.Location: Unknown 2.Device: Unrecognized	1.Score is calculated 2.If risky, login is flagged or challenged
Post Condition: Anomaly score is logged; alert triggered if needed		

Project Name: CyberLayer		Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_9		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Authentication		Test Execution date:
Test Title: Test full login and MFA authentication		
Description: Validate combined flow of login credentials and MFA verification		
Precondition: MFA must be enabled for the user		
Test Steps	Test Data	Expected Results
1.Login with valid credentials 2.Receive and input valid OTP 3.System verifies OTP and authenticates user	*Username: user@example.com *Password: Valid123 *OTP: 6-digit generated code	1.Both steps succeed 2.Session is created and logged 3.Dashboard is shown
Post Condition: User gains access to account and session starts		

Project Name: CyberLayer		Test Designed by: Ariana Rahman Lazuk
Test Case ID: TC_10		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Database		Test Execution date:
Test Title: Database update upon alert triggering		
Description: Confirm backend stores appropriate records after an alert		
Precondition : Alert rule logic must be defined		
Test Steps	Test Data	Expected Results
Simulate a suspicious login Trigger anomaly or attack alert Check database for alert log	IP: Suspicious_IP	1.Alert entry with timestamp 2.User ID and anomaly score stored
Post Condition: Alert and user session info are recorded		

Project Name: CyberLayer		Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_11		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: UI + Backend		Test Execution date:
Test Title: Verify frontend to backend communication		
Description: Ensure data entered on login UI reaches backend services correctly		
Precondition : UI form fields must be linked to API		
Test Steps	Test Data	Expected Results
1.Enter login data on UI 2.Submit and monitor backend request 3.Check data integrity on backend	*Username: testuser *Password: testpass	1.Request reaches backend 2.No data loss or corruption 3.Response returned appropriately
Post Condition: Login backend receives correct data and processes it		

Project Name: CyberLayer	Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_12	Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Dashboard	Test Execution date:

Test Title: Validate dashboard reflects login and alert activities

Description: Check that dashboard UI shows events in real-time

Precondition: Dashboard should be active and connected

Test Steps	Test Data	Expected Results
1.Perform login 2.Trigger an alert (e.g., suspicious behavior) 3.Observe dashboard updates	*Suspicious login attempt	1.Dashboard receives event and updates in real-time 2.Correct event type, time, and user info shown

Post Condition: Dashboard shows login and alert entry

Project Name: CyberLayer		Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_13		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: DDoS Detection		Test Execution date:
Test Title: Integration test for DDoS detection and response		
Description: Simulate DDoS to verify system activates alert or mitigation		
Precondition : DDoS protection module must be connected		
Test Steps	Test Data	Expected Results
1.Simulate heavy traffic 2.Monitor DDoS detection trigger 3.Check if mitigation or alert is activated	*Multiple requests from various IPs at high rate	1.Alert raised 2.Mitigation (rate limit, block) starts 3.Dashboard updated
Post Condition: DDoS event is logged and handled		

Project Name: CyberLayer	Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_14	Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Phishing detection	Test Execution date:

Test Title: Display phishing detection results on dashboard

Description: Ensure phishing detection module outputs are reflected in UI

Precondition : Detection module and dashboard must be linked

Test Steps	Test Data	Expected Results
1.Submit phishing input 2.System detects and flags it 3.Dashboard reflects detection status	*Phishing URL in input	1.Dashboard entry showing detected phishing 2.User info, time, and URL listed

Post Condition: Real-time alert visible on dashboard

Project Name: CyberLayer		Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_15		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: SQLi detection		Test Execution date:
Test Title: Test SQLi scanner on login form		
Description: Validate scanner is integrated with login input fields		
Precondition : SQLi detection logic must be active		
Test Steps	Test Data	Expected Results
1.Input SQLi payload in login field 2.Submit form 3.Check if scanner identifies attack	Input: `" ' OR ' 1 ' = ' 1 "`	1.Scanner detects injection 2.Login blocked 3.Dashboard and logs updated
Post Condition: Input blocked and scanner logs event		

Project Name: CyberLayer		Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_16		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Authentication		Test Execution date:
Test Title: Verify dashboard access after valid login		
Description: Test if user is redirected to dashboard after successful login		
Precondition : User must exist with valid credentials		
Test Steps	Test Data	Expected Results
1.Enter valid email and password 2.Click submit	Email: user@example.com Password: Valid123	1.Login successful 2.Dashboard loads with user-specific data
Post Condition: User session created; dashboard is shown		

Project Name: CyberLayer		Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_17		Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium		Test Executed by:
Module Name: Authentication		Test Execution date:
Test Title: Check email format validation		
Description: Ensure frontend validation catches incorrect email formats		
Precondition : Email field is empty or invalid		
Test Steps	Test Data	Expected Results
1.Enter invalid email format (e.g., `user#example`) 2.Click submit	Email: user#example	1.”Invalid email format" message appears 2.Login not attempted
Post Condition: Login is blocked; error displayed		

Project Name: CyberLayer		Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_18		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Authentication		Test Execution date:
Test Title: Handle login attempt with no internet		
Description: App must show meaningful error when offline		
Precondition : Internet is disconnected		
Test Steps	Test Data	Expected Results
1.Turn off internet 2.Attempt login	Email/Password: any	1."Connection error" or "Check internet connection" message 2.No crash
Post Condition: No session created; handled gracefully		

Project Name: CyberLayer		Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_19		Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium		Test Executed by:
Module Name: Authentication		Test Execution date:
Test Title: Unregistered email login		
Description: Confirm system handles non-existent emails gracefully		
Precondition : Email not present in database		
Test Steps	Test Data	Expected Results
1.Enter valid format email (not registered) 2.Enter password 3.Click submit	Email: notused@example.com	1."User not found" message displayed 2.Login blocked
Post Condition: No login; warning message shown		

Project Name: CyberLayer		Test Designed by: Monjurul Islam Nimon
Test Case ID: TC_20		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name:Authentication		Test Execution date:
Test Title: Validate expiry behavior of MFA code		
Description: Test system response when OTP/MFA expires		
Precondition: User receives OTP but waits beyond expiry time		
Test Steps	Test Data	Expected Results
1.Login with valid credentials 2.Wait until OTP expires 3.Enter expired OTP	OTP: Previously generated, now expired	1."OTP expired" message shown 2.Retry option given
Post Condition: User not logged in		

Project Name: CyberLayer	Test Designed by: Prithiraj Modak
Test Case ID: TC_21	Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Authentication	Test Execution date:

Test Title: Form submission with empty fields

Description: UI validation must prevent empty submissions

Precondition: Login page loaded

Test Steps	Test Data	Expected Results
1.Leave email and/or password empty 2.Click submit	Email: (empty) Password: (empty)	1."Required field" error messages appear 2.Login blocked

Post Condition: Submission is not sent to server

Project Name: CyberLayer	Test Designed by: Prithiraj Modak
Test Case ID: TC_22	Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name:Anomaly Detection	Test Execution date:

Test Title: Generate report for a triggered anomaly

Description: System must produce a readable report for a detected

Precondition (If any): One anomaly must be triggered for a user

Test Steps	Test Data	Expected Results
1.Trigger anomaly via suspicious login 2.Navigate to reports 3.Generate report	Event: Login from unknown IP	1.Report contains timestamp, event type, user info 2.PDF/CSV option available

Post Condition: Report stored and downloadable

Project Name: CyberLayer		Test Designed by: Prithiraj Modak
Test Case ID: TC_23		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Authentication		Test Execution date:
Test Title: Branch coverage for login validation function		
Description: Ensure all logic branches (valid, invalid, empty fields) in login validator are tested		
Precondition: Login function is implemented with conditional branches		
Test Steps	Test Data	Expected Results
1.Pass valid credentials 2.Pass invalid credentials 3.Pass empty username or password	Case 1: Valid username/password Case 2: Valid username + wrong password Case 3: Empty username	1,Proper error messages for each case 2.Code coverage includes all branches
Post Condition: Each branch executes at least once		

Project Name: CyberLayer		Test Designed by: Prithiraj Modak
Test Case ID: TC_24		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Anomaly Detection		Test Execution date:
Test Title: Condition coverage for anomaly rule logic		
Description: Validate all individual conditions in anomaly rule engine		
Precondition: Rules include multiple conditions like IP, location, time		
Test Steps	Test Data	Expected Results
1.Simulate login from new location 2.Simulate login from unknown device 3.Simulate rapid login attempts	Multiple crafted login attempts	1.Each condition triggers correct anomaly score update 2.Branches marked as covered in test reports
Post Condition: Each condition evaluated individually		

Project Name: CyberLayer		Test Designed by: Prithiraj Modak
Test Case ID: TC_25		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: SQL Injection Detection		Test Execution date:
Test Title: SQL pattern matcher decision tree coverage		
Description: Cover all decision paths used in detecting SQL injection		
Precondition: Decision tree implemented in pattern matching logic		
Test Steps	Test Data	Expected Results
1,Input various SQL payload patterns 2.Observe system detection behavior	Payloads: `' OR 1=1 --`, `DROP TABLE users`, `UNION SELECT * FROM`	1.System flags all relevant patterns correctly 2.All paths in the tree visited
Post Condition: All branches of tree exercised		

Project Name: CyberLayer	Test Designed by: Prithiraj Modak
Test Case ID: TC_26	Test Designed date:13.05.25
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Phishing Detection	Test Execution date:

Test Title: Validate phishing logic paths

Description: Exercise each possible path in phishing detection decision logic

Precondition: Phishing logic includes heuristics and pattern matching

Test Steps	Test Data	Expected Results
1.Test normal URLs 2.Test known phishing URLs 3.Test suspicious structure URLs	URLs from public phishing lists and benign samples	1.System classifies accurately 2.Full path coverage achieved in unit tests

Post Condition: All logical paths executed

Project Name: CyberLayer		Test Designed by: Prithiraj Modak
Test Case ID: TC_27		Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium		Test Executed by:
Module Name: Report Generator		Test Execution date:
Test Title: Report formatting logic branch test		
Description: Ensure all report output formats and fallback paths are exercised		
Precondition: System generates reports in multiple formats (PDF, CSV)		
Test Steps	Test Data	Expected Results
1.Generate report as PDF 2.Generate as CSV 3.Simulate unsupported format	Format: PDF, CSV, TXT	1.PDF and CSV succeed 2.TXT triggers "Unsupported format" error
Post Condition: Each logic path covered		

Project Name: CyberLayer		Test Designed by: Prithiraj Modak
Test Case ID: TC_28		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Login Session		Test Execution date:
Test Title: DB connection failure path test		
Description: Ensure system handles database failure gracefully during login		
Precondition: DB service is stopped or unreachable		
Test Steps	Test Data	Expected Results
1.Disconnect DB 2.Attempt login	Any credentials	1.“Database not available” error 2.No crash or undefined behavior
Post Condition: Error path executed; user shown proper message		

Project Name: CyberLayer		Test Designed by: Prithiraj Modak
Test Case ID: TC_29		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: MFA System		Test Execution date:
Test Title: Test MFA timeout and retry logic		
Description: Ensure MFA expiration and retry logic are covered		
Precondition: User receives valid OTP		
Test Steps	Test Data	Expected Results
1.Wait until code expires 2.Attempt to enter expired code 3.Request new code and enter it	Expired OTP, New OTP	1."Code expired" message → retry allowed → new code accepted
Post Condition: Expired code rejected; new one accepted		

Project Name: CyberLayer	Test Designed by: Md. Shahariyar Kabir Shanto
Test Case ID: TC_30	Test Designed date:13.05.25
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: System Integration	Test Execution date:
Test Title: End-to-end: Login to report generation with anomaly	
Description: Tests the full pipeline from login to report generation after anomaly	
Precondition: User must be registered with valid credentials and MFA enabled	

Test Steps	Test Data	Expected Results
1. Login with credentials 2. Enter valid MFA 3. Simulate anomaly (e.g., login from unknown device) 4. Navigate to reports and generate one	*Email: user@ex.com *Password: Valid123, OTP: 456789	1. Each step completes successfully 2. Report includes triggered anomaly

Post Condition: Anomaly logged and report successfully generated

Project Name: CyberLayer		Test Designed by: Md. Shahariyar Kabir Shanto
Test Case ID: TC_31		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: System Integration		Test Execution date:
Test Title: New user sign-up and stable login session		
Description: Test stable session lifecycle from sign-up to use		
Precondition: System must allow sign-up and login		
Test Steps	Test Data	Expected Results
1. Sign up with valid details 2. Login with those credentials 3. Use key features (e.g., view dashboard)	*Name: Test User, *Email: new@user.com	1. Account created and used without issue 2. System remains stable
Post Condition: No crash or abnormal termination		

Project Name: CyberLayer		Test Designed by: Md. Shahariyar Kabir Shanto
Test Case ID: TC_32		Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium		Test Executed by:
Module Name: Logging Module		Test Execution date:
Test Title: Log tracking of user actions		
Description: Ensure every action (login, MFA, anomaly) is logged		
Precondition: Logging service is active		
Test Steps	Test Data	Expected Results
Test Steps: 1. Login 2. Enter MFA 3. View dashboard 4. Trigger anomaly	*User session actions	1. Timestamps, user IDs, and action types properly recorded
Post Condition: Logs available in system log file		

Project Name: CyberLayer		Test Designed by: Md. Shahariyar Kabir Shanto
Test Case ID: TC_33		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Login Session		Test Execution date:
Test Title: Load test with multiple users		
Description: Ensure performance and reliability under high concurrency		
Precondition: Load testing tool or simulated users		
Test Steps	Test Data	Expected Results
1. Simulate 100+ logins concurrently	*100 user accounts	1. System remains responsive 2. All users authenticated successfully
Post Condition: No crash or dropped requests		

Project Name: CyberLayer		Test Designed by: Md. Shahariyar Kabir Shanto
Test Case ID: TC_34		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Attack Detection & Reporting		Test Execution date:
Test Title: Multi-attack simulation with unified report		
Description: Simulate DDoS, SQL Injection, and Phishing attempts		
Precondition: Attack detection modules active		
Test Steps	Test Data	Expected Results
1. Simulate 3 attack types 2. Generate report after	*Payloads for each attack	1. Each attack detected 2. Report includes time, type, and mitigation
Post Condition: Single report shows all incidents		

Project Name: CyberLayer		Test Designed by: Mst. Kamrun Nahar Omi
Test Case ID: TC_35		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: System Recovery		Test Execution date:
Test Title: Crash recovery and data integrity test		
Description: Ensure no data loss or corruption after restart		
Precondition: System is running with active user sessions		
Test Steps	Test Data	Expected Results
1. Shutdown system abruptly 2. Restart 3. Recheck session logs, reports	*Previous sessions and reports	1. All data intact 2. System stable on reboot
Post Condition: No data lost; system continues normally		

Project Name: CyberLayer		Test Designed by: Mst. Kamrun Nahar Omi
Test Case ID: TC_36		Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium		Test Executed by:
Module Name: Anomaly & Alerting		Test Execution date:
Test Title: Multiple login failures across devices		
Description: Detect abnormal failed login patterns		
Precondition: Same user attempts login from various IPs/devices		
Test Steps	Test Data	Expected Results
1. Attempt login with wrong password from 5+ locations	*IP diversity and failed logins	1. Anomaly score spikes 2. Alert logged and possibly emailed
Post Condition: System triggers alert		

Project Name: CyberLayer		Test Designed by: Mst. Kamrun Nahar Omi
Test Case ID: TC_37		Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium		Test Executed by:
Module Name: Reporting		Test Execution date:
Test Title: Export functionality check		
Description: Ensure export to PDF works correctly		
Precondition: At least one report exists		
Test Steps	Test Data	Expected Results
1. Navigate to report 2. Choose "Export to PDF"	*Any report with multiple entries	1. PDF matches screen report 2. Proper formatting and data included
Post Condition: Download completes		

Project Name: CyberLayer		Test Designed by: Mst. Kamrun Nahar Omi
Test Case ID: TC_38		Test Designed date:13.05.25
Test Priority (Low, Medium, High): High		Test Executed by:
Module Name: Detection Core		Test Execution date:
Test Title: Concurrent execution of detection modules		
Description: Ensure SQL, DDoS, and Phishing modules run without collision		
Precondition: All modules enabled		
Test Steps	Test Data	Expected Results
1. Trigger detection scenarios for all 3	*Each attack's pattern	1. No interference 2. Logs correctly recorded for each type
Post Condition: All detected without system error		

Project Name: CyberLayer		Test Designed by: Mst. Kamrun Nahar Omi
Test Case ID: TC_39		Test Designed date:13.05.25
Test Priority (Low, Medium, High): Medium		Test Executed by:
Module Name: Dashboard		Test Execution date:
Test Title: Real-time attack module status sync		
Description: Test dashboard shows updated status of all modules		
Precondition : At least one attack must be detected during the test.		
Test Steps	Test Data	Expected Results
1. Trigger SQL injection 2. Check dashboard	*Injection attempt	1. UI updates within <2 seconds 2. Correct icons/messages shown
Post Condition: Dashboard shows correct status immediately		